

Business Problem

Retail businesses generate thousands of sales transactions across different regions, product categories, customer segments, and shipping modes. Without proper analysis, it becomes difficult for management to identify sales trends, high-performing regions, customer purchasing behavior, and operational inefficiencies.

The objective of this project is to transform raw sales data into actionable business insights that support data-driven decision-making.

The goals of this analysis are to:

- Identify top-performing and underperforming regions.
- Analyze monthly sales trends to understand seasonality.
- Compare sales performance across customer segments.
- Evaluate the contribution of different shipping modes.
- Identify top-performing cities.
- Support inventory planning using quantity sold.
- Help management make strategic marketing and operational decisions.

Dataset Overview

- Row Id
- Order Id
- Order date
- Ship date
- Ship mode
- Customer Id
- Customer name
- Segment
- Country
- City
- State
- Postal code
- Region
- Product id
- Category
- Sub – Category
- Product name

- Price/Unit
- Quantity

Summary Statistics

The dataset was analyzed to understand overall business performance.

1.Total Sales

Represents the total revenue generated during the analysis period.

2.Average Sales

Shows the average revenue generated per transaction.

3.Total Transactions

Represents the total number of customer orders.

4.Total Quantity Sold

Shows the total number of products sold.

Data Validation

Before building the dashboard, the following checks were performed:

- Checked for duplicate Order IDs.
- Verified missing values.
- Standardized date formats.
- Validated sales and quantity values.
- Confirmed consistency of Region, Category, and Ship Mode.

This ensured that the dashboard was built using clean and reliable data.

Research Questions & Key Findings

1. Which Region Generates the Highest Sales?

Finding

The West region contributes the highest percentage of total sales, while the South region contributes the least.

Business Insight

The company performs exceptionally well in the West region.

Recommendation

Increase marketing efforts and customer acquisition campaigns in the South region to improve sales performance.

2. Which Shipping Mode is Preferred?

Finding

Standard Class accounts for the highest share of shipments.

Business Insight

Customers prefer economical shipping over premium delivery options.

Recommendation

Maintain efficient Standard Class logistics while promoting faster shipping services for urgent orders.

3. Which Months Generate the Highest Sales?

Finding

Sales increase significantly during September, October, and November.

Business Insight

Customer demand is seasonal, likely influenced by festive and holiday shopping.

Recommendation

Increase inventory and promotional campaigns before the peak sales season.

4. Which Cities Generate Maximum Sales?

Finding

New York generates the highest sales among all cities.

Business Insight

Large metropolitan areas contribute significantly to revenue.

Recommendation

Continue investing in high-performing cities while developing marketing strategies for lower-performing locations.

5. Which Customer Segment Contributes the Most?

Finding

The Consumer segment contributes the highest sales and order volume.

Business Insight

Individual consumers represent the largest customer base.

Recommendation

Launch loyalty programs and personalized offers to improve customer retention.

6. Which Product Category Receives the Highest Number of Orders?

Finding

Office Supplies has the highest transaction count.

Business Insight

Office Supplies are purchased frequently, indicating steady demand.

Recommendation

Ensure adequate inventory availability and optimize replenishment planning.

Business Insights

- Sales are concentrated in the West region.
- Standard Class is the most preferred shipping method.
- Consumer customers contribute the highest sales volume.
- Office Supplies are the most frequently purchased category.
- Sales peak during September to November.
- New York is the highest revenue-generating city.

Final Recommendations

- Increase marketing investment in low-performing regions.
- Prepare additional inventory before the festive sales season.
- Strengthen customer loyalty programs for Consumer customers.
- Improve logistics efficiency while maintaining Standard Class service quality.
- Expand successful strategies from top-performing cities to other markets.
- Use dashboard insights for monthly sales reviews and strategic planning.

Business Value

This dashboard enables management to:

- Monitor overall sales performance.
- Track key business KPIs in real time.
- Identify regional sales opportunities.
- Understand customer purchasing behavior.
- Improve inventory planning.
- Support faster, data-driven decision-making.

Tools & Technologies Used

- Microsoft Excel
- Pivot Tables
- Pivot Charts
- Slicers
- Data Cleaning
- Dashboard Design
- Business Analysis